Native cubic-transfer follow-up

2026-10-25 Synthetic public fixtures; partial research results

Requested scope. Prove or disprove one additional degree-three isogeny after torsion preparation; complete exceptional-chart handling; run isolated repeated end-to-end benchmarks. All three remain partial. Historical Sage evidence is preserved unchanged.

A symbolic result. In characteristic other than two or three, set $u = 3s^2$, $m = 3s$, $a = 6sv - 27s^4$ and $b = v^2 - 18s^3v + 54s^6$. For $v \neq 0$ and $v \neq 4s^3$, $P = (u, v)$ is three-torsion on $y^2 = x^3 + ax + b$. In the row basis $(1, x, y)$, its normalized translation lift is $M = N/(2v)$, with

$$N = \begin{pmatrix} mu - v & mu^2 - 3uv & 3v^2 - umv \\ -m & 2v - mu & mv \\ -1 & -u & -v \end{pmatrix}.$$

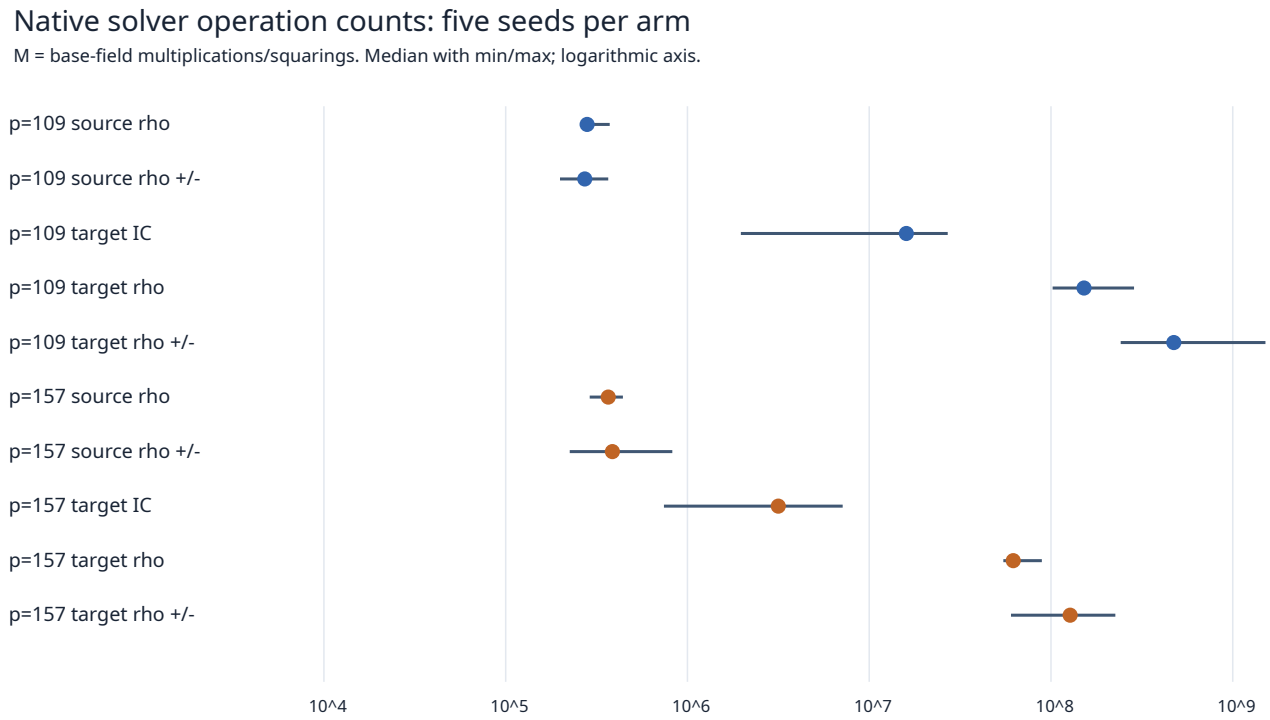
Native exact integer polynomial expansion checks $M^3 = I$, the rational translation formula after denominators are cleared, and $4a^3 + 27b^2 = 27v^3(v - 4s^3)$. The existing Rust checker replays the certificates with 64 extra points and false-identity rejection controls. The formula matches all 232 nonzero torsion actions on 29 saved fixtures. This is a rational formula for a standard theta-group construction; novelty is not claimed. It does not settle neighboring obstruction signs.

Translation certificate: `IDC1ha43e239b427420d9`.
Order certificate: `IDC1hf37d8dfb82d48731`.
Discriminant certificate: `IDC1hdfe4817e960b55ff`. These are evaluated records plus exact coefficient calculations, not Lean proofs.

Geometric chart availability. Assume a smooth plane quartic has $N_1 = p + 1$, $N_2 = p^2 + 1$, prime Jacobian order r, and $r \geq p^3 + 1 - 2p^{3/2}$. Its symmetric square has $p^2 + p + 1$ rational points. Special degree-three classes contribute at most $p + 1$ bad classes; nonreduced divisors at most $(p + 1)^2$; support on a fixed infinity line at most $4(p^2 + p + 1)$. Hence at most $B = 5p^2 + 7p + 6$ classes lack a unique reduced affine representative. For three nonzero group elements, at most $3B$ common multipliers fail. For $p \geq 19$, $r - 1 > 3B$. The positivity calculation is certificate `IDC1h6cc11bdc8a4651a3`.

This is an agent-derived geometric argument, not an independently reviewed formal proof. It does not establish that the present forward formulas or polynomial inverter work everywhere in that domain. Native projective local-jet recovery checks all 64, 568 and 1,860 rational effective degree-three divisors on the declared test curves at $p = 5, 7, 11$. It starts from support coordinates; it is not a Kummer-image inverter.

Repeated native diagnostics. Five seeds per arm and prime give 50 main runs, plus eight A/A repeats. All 58 recover correct scalars; A/A counted outputs match. A separate FLINT implementation checks all answers on the original source curves. Target points are synthetic on the fixed historical quartics and share the known answer; a native forward map has not been executed.



Run: RUN_MANIFEST.json; raw receipts: results/pinned-diagnostic/reservation.json
Target construction/transfer/inversion absent. Counts are solver-stage diagnostics.
Wall times are unqualified: missing PSI and incomplete core reservation.

p	Arm (five seeds)	Median M	Median I
109	Source rho	280,521	3,532
109	Source rho with negation	272,286	3,237
109	Target index calculus	16,001,836	667,005
109	Target rho	151,955,436	6,339,785
109	Target rho with negation	474,000,284	19,769,067
157	Source rho	366,579	4,823
157	Source rho with negation	386,379	4,583
157	Target index calculus	3,160,099	129,612
157	Target rho	61,989,685	2,567,482
157	Target rho with negation	127,470,615	5,284,059

No qualified wall times. Strict preflight refuses missing CPU and memory PSI. Diagnostic workers are pinned to CPU 3, with an exclusive lock and 16 movable threads evicted, but system threads remain eligible. No visible-tick contention is detected; this does not repair incomplete reservation or missing pressure metrics. Raw unqualified wall times, context switches, faults, RSS and A/A spread are retained.

The exact remaining mathematical obligation. Label the twelve obstructions and eligible neighbors on the relevant level-nine moduli strata. Compute the obstruction functions' divisors modulo two and the arithmetic square classes of constants. A relation $\prod u_i^{e_i} = cw^2$ excludes the all-negative star precisely when $\sum e_i$ disagrees with the character bit of c. The two saved bad centers have nonsquare neighbor products, but this does not establish a universal identity. The rational translation lift removes one interpolation dependency; neighbor labelling and modular divisor calculations remain open.

Evidence entry points. Paths are relative to this report: PROTOCOL.md; RUN_MANIFEST.json; raw reservation and job receipts; benchmark summary; FLINT scalar replay; symbolic theta audit; theta certificate replay; other certificate replay; normalization compatibility; geometric argument; square-class obligation; failed attempts and open gaps.

The source and binary hashes are in benchmark-input-sha256.txt; the worker hash begins `8c67ce79c66ac501`. Run identity is the manifest hash beginning `4dc7f9aab3f92759`. UTC process-start timestamps were not captured; filesystem times are not substituted. Compiler and host records sit in results/. No historical Sage timing is combined with these observations.

Validation and publication limits. Standalone quartic/source tests pass on both fixtures, native C++ tools build with warnings as errors, and six polynomial certificates pass the existing Rust checker. Same host/session arithmetic checks do not constitute independent-host replay. The root release-test command cannot run in the partial import checkout because Cargo.toml is absent. Conductor, Lean/Lake and five original S3 uploads remain unresolved. Full native construction, inversion, matched isolation, external comparison and calibrated accounting are not complete.

Historical component figures are unchanged: no new coverage edge was found. The scoreboard, registry, benchmark index and leaderboard are unchanged. A browsing copy of this report and graph is included under:

`benchmarks/`
`20261025-cubic-transfer-native-diagnostics/`. There is no admitted performance row and no full-pipeline speedup claim.

Prior methods. Tian, Cover Attacks for Elliptic Curves; algebraic obstruction methods, Kirschmer et al.; geometric index calculus, Diem–Thome. The native quartic arithmetic is pinned from repository PR #2170.